HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and optimized C++11+ applications for edge computing on **Linux** platforms, delivering high-performance vehicle software.
- Implemented comprehensive unit and integration tests for new features, ensuring code quality and stability within the automotive application development lifecycle.
- Collaborated with cross-functional teams, including hardware and systems engineers, to integrate vehicle applications and meet stringent production requirements.
- Designed and implemented core components for connected vehicle applications, focusing on **low-latency communication** and efficient data processing.
- Participated in **CI/CD workflows** using modern build tools like **CMake**, automating build, test, and deployment processes for faster iteration.
- Contributed to the design and implementation of automotive protocols, ensuring seamless data exchange and interoperability within the vehicle ecosystem.
- Containerized vehicle applications using **Docker** and managed deployments within a **Kubernetes** cluster, enhancing scalability and portability.

SKILLS

Languages – C++, Python, Java, SQL, R

Tools & Technologies – Linux, CMake, Bazel, Docker, Kubernetes, Git, Jenkins, Jira, Confluence, Splunk, Dynatrace

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Databases – PostgreSQL, MySQL, Oracle

Frameworks & Libraries – Spring Boot, Hibernate, JPA, pandas, NumPy, scikit-learn

Data Visualization – Power BI, Tableau, DAX, Power Query

PROJECTS

Real-Time Vehicle Data Processing Engine

- Engineered a C++ application to process real-time sensor data streams from vehicle ECUs, achieving p95 latency below **50ms**.
- Implemented data aggregation and filtering logic using a **CMake**-based build system, ensuring efficient data preparation for edge AI models.
- Integrated application components with **Linux** system services, managing resource allocation and inter-process communication for optimal performance.

Automated Build and Test Framework for Embedded Systems

- Developed a Python-based automation framework to manage the build and testing pipeline for embedded vehicle software components.
- Integrated **Jenkins** CI/CD pipelines to automatically trigger builds and run unit tests upon code commits, reducing manual deployment effort by **40%**.
- Configured test environments using **Docker** containers, ensuring consistent and reproducible test execution across different development setups.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)